PremierPreneur — Milestone 1

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Course: CIS 4394
Repo: https://github.com/jaifaburay/premierpreneur

Attached: This write-up, trace_restock_qwen.json, trace_guard.json and screenshots of the full run and the guard run.

What works
So far I have one working agent, the Business Analyst capable of running end to end on a real owner question that I crafted which was,“I have $1,000 to restock this week; what should I reorder, how many, what will it cost, what can’t you cover?” over a synthetic dataset Claude generated for me, as a 12-product store with 90 days of sales loaded from CSV into SQLite.

The agent loop is located in **my_agent.py** which follows the design of OpenAI’s open-source Codex harness with a sectioned system prompt with the role, how you work, tool guidelines, and final answer. It also has an update_plan tool, bounded retries (the same tool call with the same arguments is refused after two attempts), and a max-iteration guard of 10 as the stop condition. The model does follow the guidelines I placed where the model only chooses tools and interprets results; it never computes a number.

I had Claude craft five python tools to do the arithmetic of a sales summary, inventory status, supplier terms, a two-pass restock planner, and a discount evaluation model. Tool errors are returned to the model as observations instead of crashing the run so the model can read the error and correct itself.

Submitted run
Qwen3-8B (Hugging Face Qwen/Qwen3-8B, Q4_K_M 4-bit, via Ollama on my laptop), 5 iterations, 4 tool calls, stop_reason = final_answer. It recommended 104 Insulated Tumblers, 20 Spatula Sets and 12 Cast Iron Skillets for $999.20, listed what stayed unfunded $6,202.10 for the full target plan, skipped the overstocked Herb Garden Kit properly, and flagged P011 as unorderable because the cost is missing.

Every figure in the answer appears in a tool result. The same loop also ran on Gemini 3.8 Flash (the course’s free-tier model) on the inventory question; Gemini was returning HTTP 503 (overloaded) during today’s run window, so the local model produced the submitted trace. Setting MAX_ITERATIONS = 1 showed that the guardrails on the loop made one tool call and stopped with stop_reason = max_iterations and no answer (trace_guard.json, screenshot attached at the bottom)

What doesn't work yet:

- On iteration 3 the qwen model invented a "products" argument that doesn't exist within plan_restock. So the first version of the loop crashed on a TypeError; then the fix returns bad arguments as an error observation, and in the submitted trace the model corrects itself on the next iteration.

- On the trace.json file, the reasoning section was bypassed on the first iteration and Qwen went straight to the tool call without the one sentence justification that the prompt tells it to. While using the Gemini API, the model wrote one; which suggests that the small model follows the prompt less tightly and there's a reason for the creation of a reviewer agent. Milestone 2's evaluation set will measure it.

- In its recommendations the model stated that P002 and P004 are the "highest margin" and suggested a P008 restock but neither of these things came from a tool. The numbers were right, but the loop didn't check out the unsupported claims which is the Reviewer agent's job.

What's next (Milestone 2, due Nov 12)

- **Reviewer Agent:** check every figure and claim in the answer against the tool outputs, reject and re-ask within a retry limit.
- **Decision Agent:** which runs the conservative/moderate/aggressive planner scenarios side by side and frames the trade-off, which no single tool call does right now.
- **Planner Agent:** Making planning a required first step rather than an optional tool.
- **Gemini API Comparison:** Rerun the submitted question on Gemini for a comparison and start the 20-30 case evaluation set (missing cost, duplicate rows, an insufficient budget, bad arguments) measuring numerical accuracy, constraint compliance, latency and pass^k across models.
- **Streamlit workspace:** built last, once agent behavior is settled, so it displays finished behavior along with a CSV upload path so an owner can load their own data.
- **Real-world use cases:** Replace or supplement the synthetic store with a public Kaggle retail dataset (sales history/product catalog) so the agent is tested on real-world datasets.

AI use: Claude helped me write the data generator, the five calculation tools and the trace format, and explained each piece as I built the loop, plan tool and retry logic myself. I can explain every file in the repo.

Screenshots

Figure 1 - The full restock run on Qwen3-8B, iterations 1–5. Iteration 3 shows the invented products argument returned as an error observation; iteration 4 is the corrected call.



Figure 2 - Final answer from the same run.

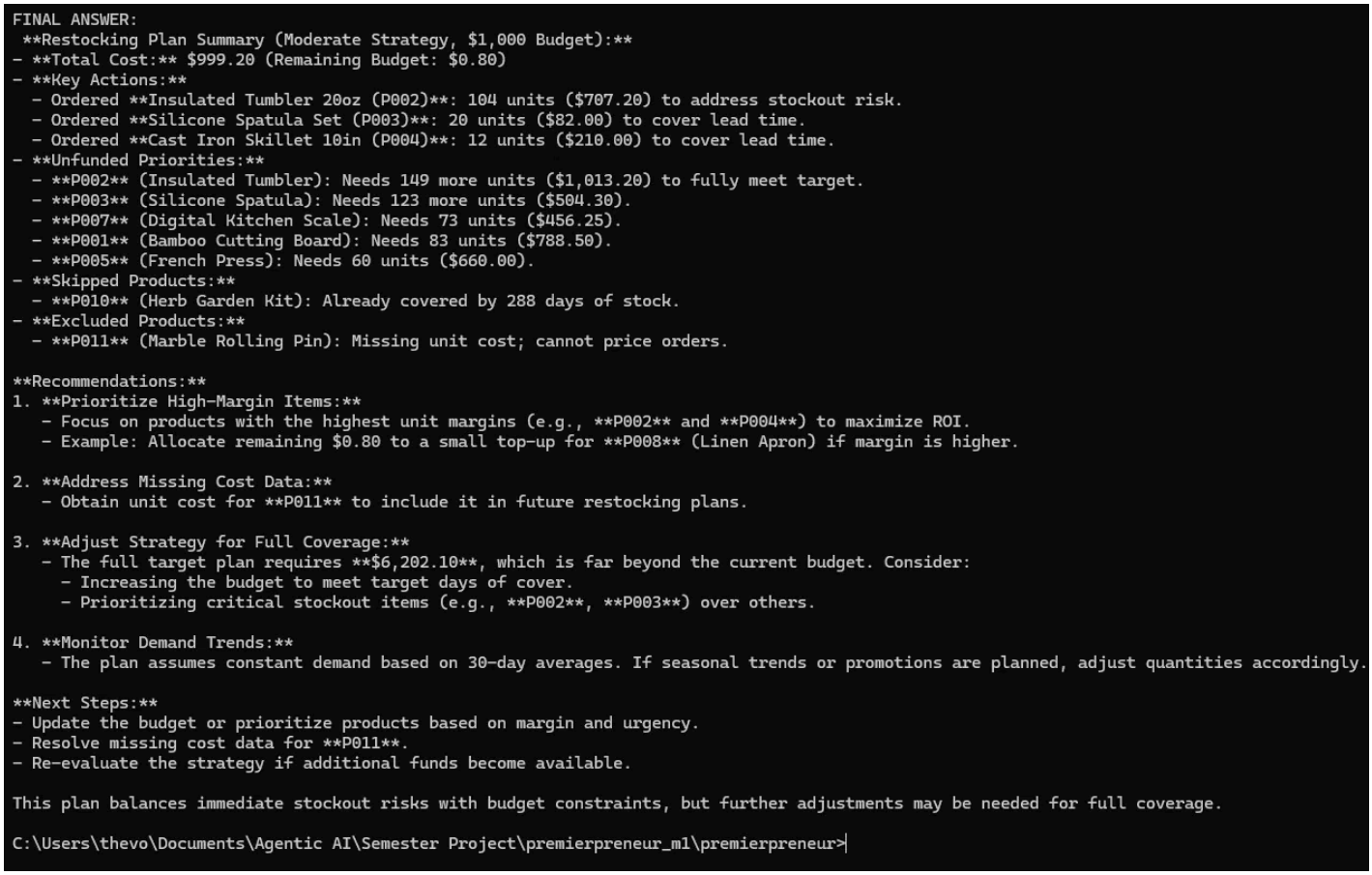


Figure 3 - Earlier version of the loop crashing with a TypeError on the same bad call, before the fix.

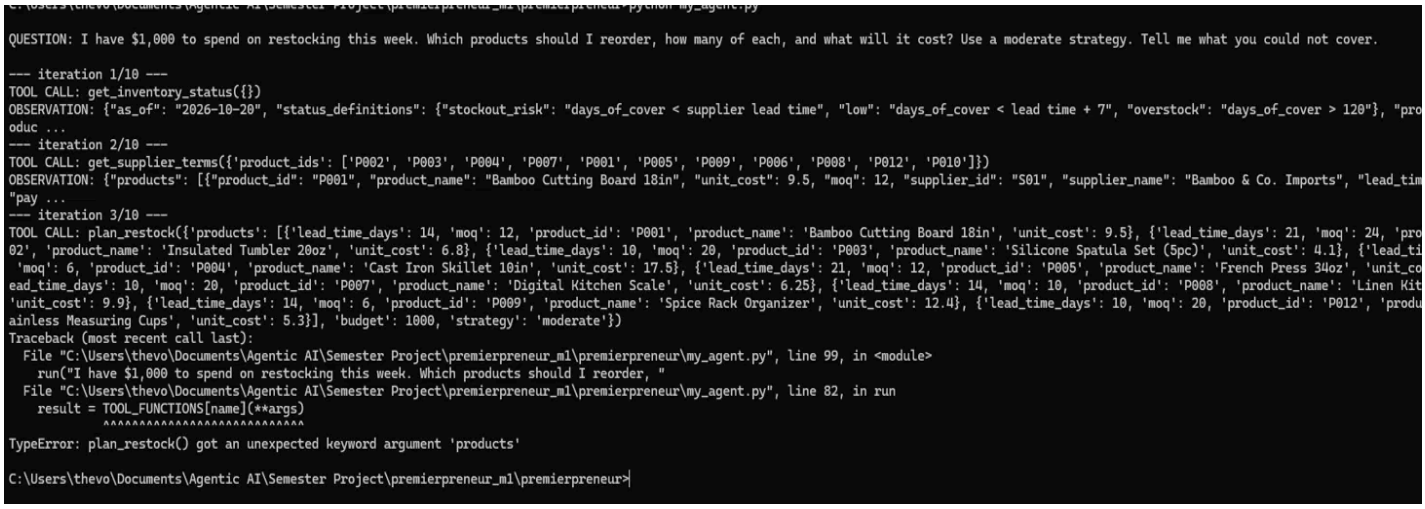


Figure 4 - The single iteration guardrails test run.

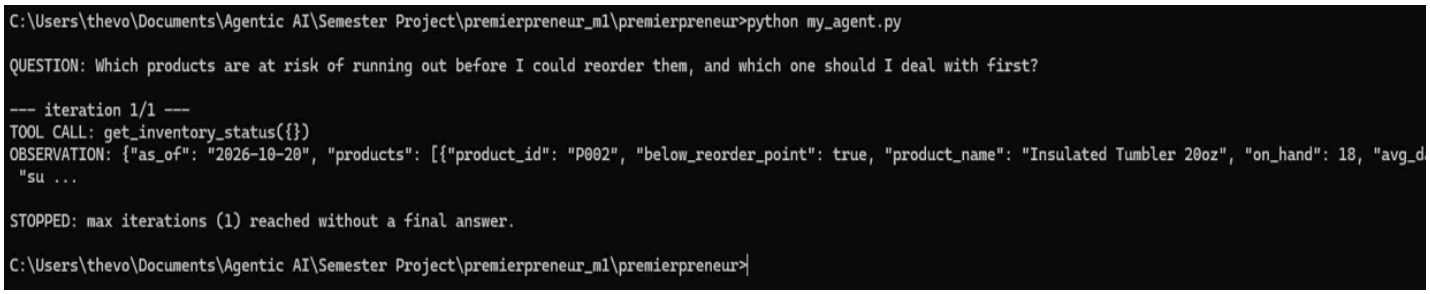


Figure 5 - Same loop on Gemini 3.8 Flash (inventory question). Unlike Qwen, Gemini writes the one-sentence justification before the tool call, as the prompt requires.

